A. Proofs of Theorem 4.1

Lemma A.1 (Principal eigenvector deviation under a spectral gap)**.** *We consider a symmetric matrix $H \in \mathbb{R}^{d\times d}$ with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \cdots$ and a spectral gap $\gamma := \lambda_1(H) - \lambda_2(H) > 0$. We define the perturbed matrix $H^+ := H + \Delta H$ where ΔH is symmetric, and we set the perturbation magnitude $s := \|\Delta H\|_{\mathrm{op}}$. We assume that the perturbation satisfies $s < \gamma$. We let $v_1 := v_1(H)$ and $v' := v_1(H^+)$ denote the unit principal eigenvectors of H and H^+, respectively, with the sign of v' chosen so that $v'^\top v_1 \geq 0$. Then we have*

$$\sin\angle(v', v_1) \leq \frac{s}{\gamma - s}, \qquad \|v' - v_1\| \leq \frac{2s}{\gamma - s}.$$

Proof. We let $u := v_1$ and $v := v'$. We decompose v into components parallel and orthogonal to u:

$$v = \alpha u + w, \qquad \alpha := u^\top v \geq 0, \qquad w \perp u, \qquad \|v\| = 1 \Rightarrow \alpha^2 + \|w\|^2 = 1.$$

We note that $\|w\| = \sin\angle(u, v)$.

We let $\mu_1 := \lambda_1(H^+)$ so that $(H + \Delta H)v = \mu_1 v$. Rearranging the terms, we obtain

$$(H - \mu_1 I)v = -\Delta H\, v.$$

We project onto $u^\perp$ using $P_\perp := I - uu^\top$. Since $P_\perp u = 0$ and $P_\perp v = w$, we arrive at

$$(H - \mu_1 I)w = -P_\perp \Delta H\, v.$$

Taking norms and using $\|P_\perp\| \leq 1$, we obtain

$$\|(H - \mu_1 I)w\| \leq \|\Delta H\|_{\mathrm{op}}\|v\| = s. \tag{A.1}$$

Next, we lower bound the left-hand side. By applying Weyl's inequality, we have $\mu_1 = \lambda_1(H^+) \geq \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1 - s$, so that

$$\mu_1 - \lambda_2 \geq (\lambda_1 - s) - \lambda_2 = \gamma - s. \tag{A.2}$$

Since $w \perp u$ lies in the invariant subspace corresponding to eigenvalues at most λ_2, we observe that the smallest eigenvalue of $(\mu_1 I - H)$ restricted to $u^\perp$ is at least $\mu_1 - \lambda_2$. Hence, we have

$$\|(\mu_1 I - H)w\| \geq (\mu_1 - \lambda_2)\|w\| \geq (\gamma - s)\|w\|. \tag{A.3}$$

Combining (A.1) and (A.3), we find $(\gamma - s)\|w\| \leq s$, i.e.,

$$\|w\| \leq \frac{s}{\gamma - s}.$$

Therefore, we conclude $\sin\angle(v', v_1) = \|w\| \leq s/(\gamma - s)$. Finally, because $v'^\top v_1 \geq 0$, we note that the angle $\theta := \angle(v', v_1) \in [0, \pi/2]$ and

$$\|v' - v_1\| = 2\sin(\theta/2) \leq 2\sin\theta = 2\|w\| \leq \frac{2s}{\gamma - s}.$$

This completes the proof. □

We consider the local geometry of a thrice-differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. Let $H := \nabla^2 f(x)$ be the Hessian at x, with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \ldots$. Given an update step $\Delta x = \eta\tilde{h}$, the perturbed Hessian is $H^+ := \nabla^2 f(x + \Delta x)$, and we define the residual operator $\Delta H := H^+ - H$.

Definition A.2 (Spectral and Smoothness Parameters)**.** The following quantities characterize the stability of the system:

- **Eigengap:** $\gamma := \lambda_1(H) - \lambda_2(H) > 0$.
- **Perturbation Magnitude:** $s := \|\Delta H\|_{\mathrm{op}}$.

- **Uniform Third-derivative Bound:** A constant $M_3 > 0$ such that:

$$|\nabla^3 f(z)[a,b,c]| \le M_3 \|a\| \|b\| \|c\|, \quad \forall a,b,c \in \mathbb{R}^d.$$

Assumption 7 (Decrease Conditions)**.** *To guarantee the strict decrease of the top eigenvalue, we assume:*

- ***Non-degeneracy:*** *The perturbation is bounded by $s < \gamma$.*
- ***Descent Direction (C1):*** $v_1^\top \Delta H v_1 \le -\eta\kappa$ *for some* $\kappa > 0$.
- ***Step Size Sufficiency (C2):*** $\eta\kappa > C_0 s$, *where* $C_0 := \frac{4(\|H\|_{\mathrm{op}}+s)}{\gamma - s}$ *is the spectral sensitivity coefficient.*

Theorem A.3 (Strict Decrease of λ_1)**.** *Under Assumption 7, the perturbation magnitude satisfies $s \le M_3 \eta \|\tilde{h}\|$, and the leading eigenvalue of the perturbed Hessian satisfies:*

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}\, s. \tag{13}$$

Proof. We apply the fundamental theorem of calculus to the matrix-valued function $t \mapsto \nabla^2 f(x + t\Delta x)$ to write

$$\Delta H = \int_0^1 \frac{d}{dt} \nabla^2 f(w + t\Delta x)\, dt = \int_0^1 \nabla^3 f(w + t\Delta w)[\,\cdot, \cdot, \Delta w\,]\, dt. \tag{14}$$

Consequently, for any unit vectors u, v, we bound the operator norm s as follows:

$$\begin{aligned} s := \|\Delta H\|_{\mathrm{op}} &= \sup_{\|u\|=\|v\|=1} \left| u^\top \Delta H\, v \right| \\ &= \sup_{\|u\|=\|v\|=1} \left| \int_0^1 \nabla^3 f(x + t\Delta x)[u, v, \Delta x]\, dt \right| \\ &\le \sup_{\|u\|=\|v\|=1} \int_0^1 M_3 \|u\|\, \|v\|\, \|\Delta x\|\, dt \\ &= M_3 \|\Delta x\| = M_3\, \eta\, \|\tilde{h}\|. \end{aligned} \tag{15}$$

Using the Rayleigh quotient, we express the leading eigenvalue as $\lambda_1(H + \Delta H) = {v'}^\top (H + \Delta H)\, v'$. By adding and subtracting $v_1^\top (H + \Delta H)\, v_1$, we decompose this term as

$$\lambda_1(H + \Delta H) = v_1^\top (H + \Delta H)\, v_1 + \Big({v'}^\top (H + \Delta H)\, v' - v_1^\top (H + \Delta H)\, v_1 \Big). \tag{16}$$

Moreover, utilizing the eigen-equation $Hv_1 = \lambda_1(H) v_1$ and the normalization $\|v_1\| = 1$, we simplify the first term:

$$v_1^\top (H + \Delta H)\, v_1 = \lambda_1(H) + v_1^\top \Delta H\, v_1. \tag{17}$$

We recall that for any symmetric matrix A and unit vectors u, v, the inequality $\left| u^\top A u - v^\top A v \right| \le 2\|A\|_{\mathrm{op}} \|u - v\|$ holds. Applying this bound to our context, we obtain

$$\begin{aligned} \left| {v'}^\top (H + \Delta H) v' - v_1^\top (H + \Delta H) v_1 \right| &\le 2\|H + \Delta H\|_{\mathrm{op}} \|v' - v_1\| \\ &\le 2(\|H\|_{\mathrm{op}} + \|\Delta H\|_{\mathrm{op}}) \|v' - v_1\| \\ &= 2(\|H\|_{\mathrm{op}} + s) \|v' - v_1\|. \end{aligned} \tag{18}$$

To estimate the eigenvector difference, we set $u := v_1$ and $v := v'$. We decompose v as

$$v = \alpha u + w, \qquad \alpha = u^\top v \ge 0, \qquad w \perp u, \qquad \alpha^2 + \|w\|^2 = 1.$$

We define $\mu_1 := \lambda_1(H + \Delta H)$. Since $(H + \Delta H) v = \mu_1 v$, we rearrange the equation to

$$(H - \mu_1 I) v = -\Delta H\, v. \tag{19}$$

We introduce $P^\perp := I - uu^\top$ as the orthogonal projector onto $u^\perp$. Applying $P^\perp$ to (19) and utilizing the properties $P^\perp u = 0$ and $P^\perp v = w$, we derive

$$(H - \mu_1 I)w = -P^\perp \Delta H\, v. \tag{20}$$

Taking the norms, we obtain

$$\|(H - \mu_1 I)w\| \le \|P^\perp\|_{\mathrm{op}} \|\Delta H\|_{\mathrm{op}} \|v\| \le s, \tag{21}$$

where we used $\|P^\perp\|_{\mathrm{op}} = 1$ and $\|v\| = 1$. Next, by invoking Weyl's inequality, we have

$$\mu_1 = \lambda_1(H + \Delta H) \ge \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1(H) - s, \tag{22}$$

and hence

$$\mu_1 - \lambda_2(H) \ge (\lambda_1(H) - s) - \lambda_2(H) = \gamma - s. \tag{23}$$

Since $w \perp u$, the vector w lies in the orthogonal complement of the top eigenspace of H. Therefore, we lower bound the norm as

$$\|(\mu_1 I - H)w\| \ge (\mu_1 - \lambda_2(H))\|w\| \ge (\gamma - s)\|w\|. \tag{24}$$

We note that $\|(H - \mu_1 I)w\| = \|(\mu_1 I - H)w\|$. Combining (21) and (24), we find that

$$(\gamma - s)\|w\| \le s \quad \Rightarrow \quad \|w\| \le \frac{s}{\gamma - s}.$$

Finally, since $\alpha \ge 0$ (implying the angle between u and v lies in $[0, \pi/2]$), we have $\|v - u\| \le 2\|w\|$, and thus we conclude

$$\|v' - v_1\| = \|v - u\| \le \frac{2s}{\gamma - s}. \tag{25}$$

Substituting (17) into (16) and utilizing the bounds (18) and (25), we obtain

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + 2(\|H\|_{\mathrm{op}} + s) \cdot \frac{2s}{\gamma - s}.$$

This can be rewritten as

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \underbrace{\frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}}_{=:C_0} s. \tag{26}$$

If condition (C1) holds, i.e., $v_1^\top \Delta H v_1 \le -\eta\kappa$, then (26) implies $\lambda_1(H + \Delta H) \le \lambda_1(H) - \eta\kappa + C_0 s$. If in addition (C2) is satisfied, i.e., $\eta\kappa > C_0 s$, then we strictly obtain

$$\lambda_1(H + \Delta H) < \lambda_1(H), \tag{27}$$

which proves the desired strict decrease. □

B. Proofs of Theorem 4.2

Definition B.1 (Algorithm Quantities)**.** We define the following quantities to characterize the convergence rate and perturbation bias:

- **Gradient Notations:** Let $g_t := \nabla f(w_t)$ be the true gradient, $h_t := \nabla f(w_t + \rho\epsilon_t)$ be the perturbed gradient, and $\hat{h}_t$ be the stochastic estimate.
- **SAM Bias:** $D_\rho := \beta^2\rho^2(1 + C_1\alpha^2)$, representing the error upper bound $\mathbb{E}\|g_t - h_t\|^2 \le D_\rho$.
- **Descent Coefficient:** $c_0 := \frac{3}{4} - 2\alpha^2$, which serves as the effective curvature scaling.

Assumption 8 (Stochastic Setup and Parameters)**.** *We adopt the following standard assumptions:*

1. ***Oracle Properties:*** *The stochastic gradient is unbiased, $\mathbb{E}[\hat{h}_t | w_t] = h_t$, with bounded variance $\mathbb{E}[\|\hat{h}_t - h_t\|^2] \le \sigma^2$.*

2. ***Perturbation Bound:*** *The noise satisfies* $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$.

3. ***Parameter Selection:*** *We choose* α *such that* $\alpha^2 < 3/8$ *(ensuring* $c_0 > 0$*) and set the step size as:*

$$\eta = \min\left\{\frac{1}{2\beta}, \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}, \quad \textit{where } \Delta := f(w_0) - f_*.$$

Theorem B.2 (Convergence Rate)**.** *Under Assumption 8, the X-SAM algorithm guarantees the following bound on the average squared gradient norm:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \;\le\; c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)D_\rho, \tag{28}$$

where $c_1, c_2 > 0$ *are constants depending only on the descent coefficient* c_0.

Proof. We assume $\alpha \in [0,2]$ and $\|v_t\| = 1$. Consequently, for the projector $P_t := I - \alpha v_t v_t^\top$, we have the operator norm bound $\|P_t\|_{\mathrm{op}} = \max\{1, |1-\alpha|\} \le 1$. Let us define the gradient terms: $\|P_t\|_{\mathrm{op}} = \max\{1, |1-\alpha|\} \le 1$. Let

$$g_t := \nabla f(w_t), \qquad h_t := \nabla f(w_t + \rho\epsilon_t), \qquad \hat{h}_t := \nabla f(w_t + \rho\epsilon_t; \xi_t), \qquad \tilde{h}_t := P_t\hat{h}_t,$$

and the update rule $w_{t+1} = w_t - \eta\tilde{h}_t$. We assume that the stochastic gradient is unbiased with bounded variance, i.e.,

$$\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t] = h_t, \qquad \mathbb{E}\Big[\|\hat{h}_t - h_t\|^2 \mid w_t, \epsilon_t, v_t\Big] \le \sigma^2.$$

By the tower property, this implies the global statements

$$\mathbb{E}\langle u_t, \hat{h}_t - h_t\rangle = 0 \;\; (\forall\, u_t = u_t(w_t, \epsilon_t, v_t)), \qquad \mathbb{E}\|\hat{h}_t - h_t\|^2 \le \sigma^2. \tag{29}$$

Assume $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$ for some $C_1 > 0$. Since ∇f is β-Lipschitz, we have the pointwise bound

$$\|g_t - h_t\| = \big\|\nabla f(w_t) - \nabla f(w_t + \rho\epsilon_t)\big\| \le \beta\rho\|\epsilon_t\| = \beta\rho \tag{30}$$

By the β-smoothness of f, we have

$$f(w_{t+1}) \le f(w_t) + \langle g_t, w_{t+1} - w_t\rangle + \frac{\beta}{2}\|w_{t+1} - w_t\|^2 = f(w_t) - \eta\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\|\tilde{h}_t\|^2.$$

Taking the total expectation yields

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\mathbb{E}\|\tilde{h}_t\|^2. \tag{31}$$

Bounding the inner product term: Since $\tilde{h}_t = P_t\hat{h}_t$, using the tower property and the definition of P_t, we derive:

$$\begin{aligned}
\mathbb{E}\langle g_t, \tilde{h}_t\rangle &= \mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle\big] = \mathbb{E}\Big[\mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle \mid w_t, \epsilon_t, v_t\big]\Big] \\
&= \mathbb{E}\Big[\big\langle g_t, P_t\,\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t]\big\rangle\Big] = \mathbb{E}\langle g_t, P_t h_t\rangle \\
&= \mathbb{E}\langle g_t, h_t\rangle - \alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle]\,.
\end{aligned} \tag{32}$$

Since $\|P_t\|_{\mathrm{op}} \le 1$, we have $\|\tilde{h}_t\| = \|P_t\hat{h}_t\| \le \|\hat{h}_t\|$ pointwise. Consequently, this yields the inequality:

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|\hat{h}_t\|^2. \tag{33}$$

We decompose the stochastic gradient as $\hat{h}_t = h_t + (\hat{h}_t - h_t)$ and apply (29) to derive:

$$\begin{aligned}
\mathbb{E}\|\hat{h}_t\|^2 &= \mathbb{E}\langle \hat{h}_t, \hat{h}_t\rangle \\
&= \mathbb{E}\Big\langle h_t + (\hat{h}_t - h_t),\, h_t + (\hat{h}_t - h_t)\Big\rangle \\
&= \mathbb{E}\Big[\|h_t\|^2 + 2\langle h_t, \hat{h}_t - h_t\rangle + \|\hat{h}_t - h_t\|^2\Big] \\
&= \mathbb{E}\|h_t\|^2 + 2\,\mathbb{E}\langle h_t, \hat{h}_t - h_t\rangle + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&= \mathbb{E}\|h_t\|^2 + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&\le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (34)
\end{aligned}$$

Combining (33) and (34), we deduce that

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (35)$$

Substituting (32) and (35) into (31), we arrive at

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, h_t\rangle + \eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] + \frac{\beta\eta^2}{2}\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (36)$$

We utilize the polarization identity $2\langle a, b\rangle = \|a\|^2 + \|b\|^2 - \|a-b\|^2$ (pointwise) and take the expectation to write

$$-\eta\,\mathbb{E}\langle g_t, h_t\rangle = -\frac{\eta}{2}\mathbb{E}\|g_t\|^2 - \frac{\eta}{2}\mathbb{E}\|h_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2. \qquad (37)$$

For any $\delta \in (0,1)$, we apply Young's inequality using the fact that $|v_t| = 1$ to obtain the pointwise bound:

$$\begin{aligned}
\alpha\langle h_t, v_t\rangle\langle g_t, v_t\rangle &= \Big(\alpha\langle h_t, v_t\rangle\Big)\Big(\langle g_t, v_t\rangle\Big) \\
&\le \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{1}{2\delta}\Big(\alpha\langle h_t, v_t\rangle\Big)^2 \qquad (\forall\,\delta > 0) \qquad (38) \\
&= \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{\alpha^2}{2\delta}\,\langle h_t, v_t\rangle^2 \\
&\le \frac{\delta}{2}\,\|g_t\|^2 + \frac{\alpha^2}{2\delta}\,\|h_t\|^2, \qquad (39)
\end{aligned}$$

Hence, we have

$$\eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] \le \frac{\eta\delta}{2}\mathbb{E}\|g_t\|^2 + \frac{\eta\alpha^2}{2\delta}\mathbb{E}\|h_t\|^2. \qquad (40)$$

Substituting (37) and (40) into (36), we obtain

$$\begin{aligned}
\mathbb{E}f(w_{t+1}) \le\ & \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2 \\
&+ \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (41)
\end{aligned}$$

From the triangle inequality $\|h_t\| \le \|g_t\| + \|h_t - g_t\|$, we have

$$\|h_t\|^2 \le 2\|g_t\|^2 + 2\|h_t - g_t\|^2. \qquad (42)$$

Taking the expectation and using (30) (which implies $\|h_t - g_t\|^2 = \|g_t - h_t\|^2 \le D_\rho$ point-wise), we obtain

$$\mathbb{E}\|h_t\|^2 \le 2\,\mathbb{E}\|g_t\|^2 + 2D_\rho, \qquad \mathbb{E}\|g_t - h_t\|^2 \le D_\rho. \qquad (43)$$

We fix $\delta = \frac{1}{2}$, $\beta\eta \le \frac{1}{2}$, and $\alpha \le \sqrt{\frac{3}{8}}$, so that we satisfy the descent condition

$$\frac{\alpha^2}{\delta} + \beta\eta \le \frac{3-\delta}{2} \qquad (44)$$

Applying (43), we obtain

$$\begin{aligned}\mathbb{E}f(w_{t+1}) &\le \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}D_\rho + \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\Big(2\,\mathbb{E}\|g_t\|^2 + 2D_\rho\Big) + \frac{\beta\eta^2}{2}\sigma^2 \\ &\le \mathbb{E}f(w_t) - \eta\Big(\frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta\Big)\mathbb{E}\|g_t\|^2 + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2 \qquad (45)\end{aligned}$$

We define

$$c := \frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta.$$

We fix $\delta = \frac{1}{2}$ and assume $\beta\eta \le \frac{1}{2}$ and $\alpha^2 < \frac{3}{8}$. Then for this c we have

$$c = \frac{5}{4} - 2\alpha^2 - \beta\eta \ge \frac{5}{4} - 2\alpha^2 - \frac{1}{2} = \frac{3}{4} - 2\alpha^2 =: c_0 > 0,$$

which implies $1/c \le 1/c_0$.

From (45) we directly obtain

$$\eta c\,\mathbb{E}\|g_t\|^2 \le \mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1}) + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2, \qquad (46)$$

$$\begin{aligned}\eta c\sum_{t=0}^{T-1}\mathbb{E}\|g_t\|^2 &\le \sum_{t=0}^{T-1}\big(\mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1})\big) + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &= \mathbb{E}\big[f(w_0) - f(w_T)\big] + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &\le \Delta + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2,\end{aligned}$$

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \qquad (47)$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \qquad (48)$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho \le (1 + 2\alpha^2)\,D_\rho.$$

Moreover, by the definition of η, we have

$$\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \qquad \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}.$$

We choose

$$\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}.$$

Plugging these bounds into (48), we obtain that there exist constants $c_1, c_2 > 0$ (depending only on c_0) such that

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1 + 2\alpha^2)\,D_\rho, \qquad D_\rho = \beta^2\rho^2. \qquad (49)$$

In particular, we note that when $\sigma^2 > 0$ the stochastic term scales as $O(1/\sqrt{T})$, and when $\sigma = 0$ the bound reduces to $O(1/T)$ up to the perturbation bias term proportional to D_ρ.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \tag{50}$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{51}$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\tfrac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho \le (1+2\alpha^2)\,D_\rho. \tag{52}$$

Substituting (52) into (48), we obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{53}$$

We choose the step size $\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}$. Then $\beta\eta \le \frac{1}{2}$ and, moreover, $\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}$. We obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\left(2\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}}$$

$$= \frac{2}{c_0}\frac{\beta\Delta}{T} + \frac{3}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho \tag{54}$$

$$\le c_1\left(\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)\,D_\rho.$$

□

Theorem B.3 (First-order change of the leading eigenvalue)**.** *We assume that $f:\mathbb{R}^d\to\mathbb{R}$ is $\mathcal{C}^4$ in a neighborhood of w_t, and we define $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We assume the existence of $\rho>0$ such that the third and fourth derivatives are locally bounded: $M_3 := \sup_{\|u-w_t\|\le\rho}\|\nabla^3 f(u)\|_{\mathrm{op}} < \infty$, $M_4 := \sup_{\|u-w_t\|\le\rho}\|\nabla^4 f(u)\|_{\mathrm{op}} < \infty$. We further assume that H_t possesses a simple largest eigenvalue $\lambda_1(w_t)$ with unit eigenvector $v_1(w_t)$, and we define the spectral gap as $\gamma := \lambda_1(w_t)-\lambda_2(w_t) > 0$. We consider the update $\Delta w := w_{t+1}-w_t = -\eta g_t$ with $\eta>0$ chosen sufficiently small to ensure that $\|H(w_t+\Delta w)-H(w_t)\|_{\mathrm{op}} < \gamma/2$. We additionally assume that $\|(I-v_1v_1^\top)\nabla\lambda_1(w_t)\| \le \varepsilon$, and$v_1(w_t)^\top g_t = 0$. Under these assumptions, we establish that the leading eigenvalue increment $\Delta\lambda_1 := \lambda_1(w_{t+1})-\lambda_1(w_t)$ satisfies*

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on M_3, M_4 and on the universal spectral constant in the perturbation remainder.

Proof. We let $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We define the updates as

$$\Delta w := w_{t+1}-w_t = -\eta g_t, \qquad \Delta H := H(w_t+\Delta w)-H(w_t).$$

By invoking the *analytic perturbation theorem* (Kato, 1966), we assert that there exists a neighborhood of H_t (in particular, whenever $\|\Delta H\|_{\mathrm{op}} < \gamma/2$) on which $\lambda_1(\cdot)$ is differentiable and

$$D\lambda_1(H_t)[E] = v_1^\top E\, v_1.$$

Hence, the first-order expansion yields

$$\Delta\lambda_1 := \lambda_1(w_{t+1}) - \lambda_1(w_t) = v_1^\top(\Delta H)\, v_1 + r_{\text{spec}}, \tag{55}$$

with the gap-controlled remainder satisfying

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma}. \tag{56}$$

We let $\mathcal{B}$ be a ball around w_t containing the segment $\{w_t + s\Delta w : s \in [0,1]\}$, and we define the Lipschitz constants

$$L_0 := \sup_{w\in\mathcal{B}} \|DH(w)\|_{\text{op}}, \qquad L_1 := \sup_{w\in\mathcal{B}} \|D^2H(w)\|_{\text{op}}.$$

(In particular, if $\|\nabla^3 f(w)\|$ is bounded on $\mathcal{B}$ by M_3, then $L_0 \le M_3$. If moreover $\|\nabla^4 f(w)\|$ is bounded on $\mathcal{B}$ by M_4, then $L_1 \le M_4$.) Consequently, we bound the perturbation norm as

$$\|\Delta H\|_{\text{op}} = \|H(w_t + \Delta w) - H(w_t)\|_{\text{op}} \le L_0\|\Delta w\|. \tag{57}$$

We define $\phi(s) := H(w_t + s\Delta w)$. Then, noting that $\phi'(s) = DH(w_t + s\Delta w)[\Delta w]$, we write

$$\Delta H = \int_0^1 DH(w_t + s\Delta w)[\Delta w]\, ds. \tag{58}$$

Adding and subtracting $DH(w_t)[\Delta w]$, we obtain

$$\Delta H = DH(w_t)[\Delta w] + R_H, \qquad R_H := \int_0^1 \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w]\, ds. \tag{59}$$

In coordinates, we express the differential as

$$DH(w_t)[\Delta w] = \sum_{i=1}^{d} \frac{\partial H}{\partial w_i}(w_t)\, \Delta w_i =: \nabla_w H[w_t] \cdot \Delta w. \tag{60}$$

Since D^2H is bounded by L_1 on $\mathcal{B}$, the map DH is Lipschitz on $\mathcal{B}$ with constant L_1, i.e.,

$$\|DH(u) - DH(w)\|_{\text{op}} \le L_1\|u - w\| \qquad \forall\, u, w \in \mathcal{B}.$$

Therefore, for all $s \in [0,1]$, we have

$$\|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}} \le L_1\|s\Delta w\| = sL_1\|\Delta w\|.$$

Substituting this bound into the definition of R_H, we derive

$$\begin{aligned}
\|R_H\|_{\text{op}} &\le \int_0^1 \Big\| \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w] \Big\|_{\text{op}}\, ds \\
&\le \int_0^1 \|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}}\, \|\Delta w\|\, ds \\
&\le \int_0^1 sL_1\|\Delta w\|^2\, ds = \tfrac{1}{2}L_1\|\Delta w\|^2.
\end{aligned} \tag{61}$$

Combining (55) and (59), we arrive at

$$\Delta\lambda_1 = v_1^\top\big(DH(w_t)[\Delta w]\big)v_1 \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}. \tag{62}$$

Using $\|v_1\| = 1$ and (61), we find

$$|v_1^\top R_H v_1| \le \|R_H\|_{\text{op}} \le \tfrac{1}{2}L_1\|\Delta w\|^2.$$

Moreover, by applying (56) and (57), we obtain

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma} \le C_{\text{spec}} \frac{L_0^2}{\gamma} \|\Delta w\|^2.$$

Expanding $DH(w_t)[\Delta w]$ using (60) gives

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \, v_1^\top \frac{\partial H}{\partial w_i}(w_t) v_1.$$

We let $H_t := H(w_t)$ and suppose the top eigenpair (λ_1, v_1) satisfies $H_t v_1 = \lambda_1 v_1, \qquad \|v_1\|_2 = 1$. We differentiate both sides with respect to the coordinate w_i (evaluated at w_t). Using the product rule, we obtain

$$\frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1 \;+\; \lambda_1 \frac{\partial v_1}{\partial w_i}. \tag{63}$$

Multiplying by $v_1^\top$ from the left, we obtain

$$v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1^\top v_1 \;+\; \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}. \tag{64}$$

Since H_t is symmetric, we note that

$$v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}, \tag{65}$$

so these terms cancel on both sides. Therefore, we derive

$$\frac{\partial \lambda_1}{\partial w_i}(w_t) = v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1, \qquad i = 1, \ldots, d. \tag{66}$$

and hence

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \frac{\partial \lambda_1}{\partial w_i}(w_t) = \langle \nabla \lambda_1(w_t),\, \Delta w \rangle. \tag{67}$$

Substituting (67) into (62) and using $\Delta w = -\eta g_t$, we obtain

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}.$$

Using the bounds above and the relation $\|\Delta w\| = \eta \|g_t\|$, we have

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; O\big(\eta^2 \|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2 \|g_t\|^2}{\gamma}\Big),$$

where the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}.

We decompose the gradient into the component orthogonal to v_1 and the component parallel to v_1:

$$\nabla \lambda_1(w_t) = (I - v_1 v_1^\top) \nabla \lambda_1(w_t) + v_1 v_1^\top \nabla \lambda_1(w_t). \tag{68}$$

Taking the inner product with Δw, we obtain

$$\langle \nabla \lambda_1(w_t), \Delta w \rangle = \langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle + \langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \rangle \tag{69}$$

By applying the Cauchy–Schwarz inequality, we have

$$\big|\langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle\big| \le \|(I - v_1 v_1^\top) \nabla \lambda_1(w_t)\| \, \|\Delta w\| \le \varepsilon \|\Delta w\|. \tag{70}$$

Moreover, using the identity $v_1 v_1^\top x = (v_1^\top x)\, v_1$ for any vector x, we observe that

$$v_1 v_1^\top \nabla \lambda_1(w_t) = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, v_1, \tag{71}$$

$$\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, \big(v_1^\top \Delta w\big), \tag{72}$$

$$\big|\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle\big| \le \big|v_1^\top \nabla \lambda_1(w_t)\big| \, \big|v_1^\top \Delta w\big|. \tag{73}$$

Therefore, in the orthogonal case where $v_1^\top g_t = 0$, we deduce

$$\big|\eta\langle \nabla\lambda_1(w_t), g_t\rangle\big| = \big|\langle \nabla\lambda_1(w_t), \Delta w\rangle\big| \le \varepsilon\|\Delta w\| = \eta\varepsilon\|g_t\|.$$

Finally, we obtain the single-step eigenvalue change bound

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}. Hence, we conclude that when ε is sufficiently small, the first-order contribution is suppressed and λ_1 changes only by higher-order terms in one step. □

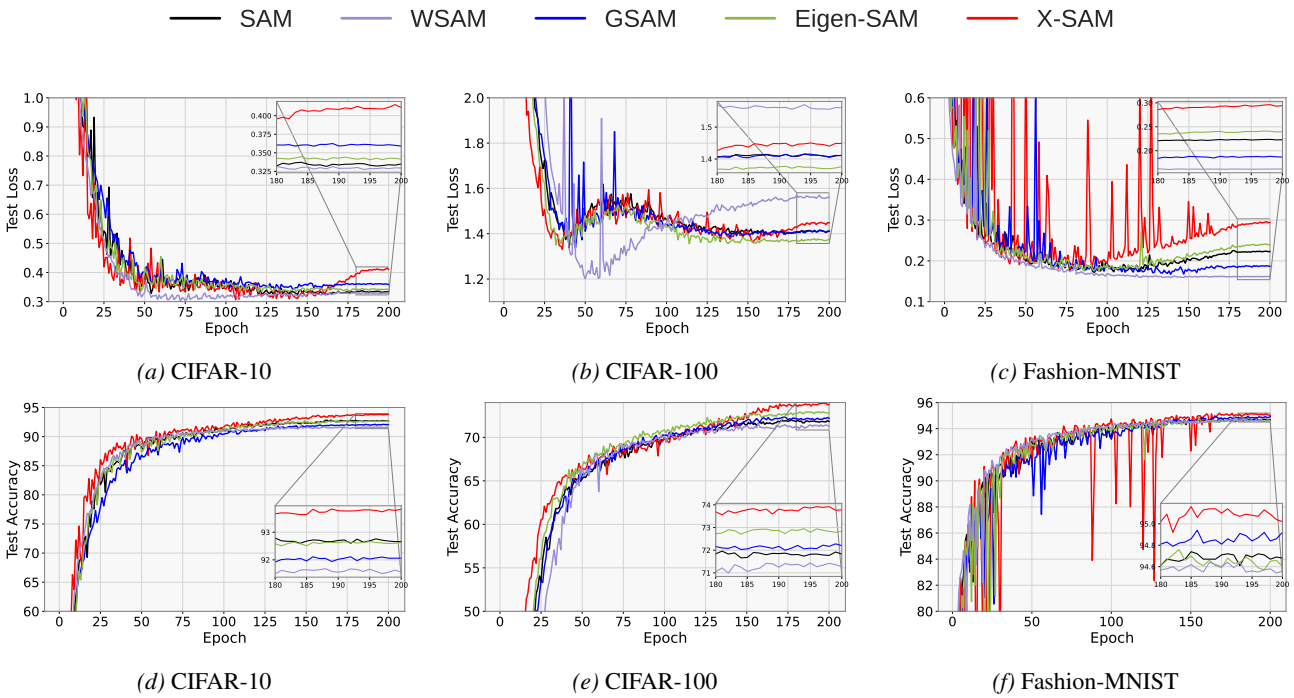
C. Additional Experimental Results

In this appendix, we provide a comprehensive empirical evaluation of X-SAM through ten specialized experiments to further validate its effectiveness and robustness. We first analyze the training dynamics and generalization performance through test loss and accuracy curves (C.1), evaluations on the Tiny ImageNet dataset (C.2), and ViT-Tiny model fine-tuning (C.8), while also demonstrating efficiency via test accuracy under equal training time (C.9). To assess the stability of X-SAM, we conduct extensive sensitivity investigations on key hyperparameters, including the perturbation radius ρ (C.3), the gradient correction coefficient α (C.4), and the parameters p (C.6)and q (C.7). Finally, we explore the underlying optimization mechanism by examining the Hessian spectrum (C.5) to visualize eigenvalue distribution and monitoring the parameter norm (C.10) to characterize the implicit regularization and flatness found by X-SAM.

C.1. Evaluation on Test Loss and Accuracy

We evaluated the test accuracy and test loss of various methods using the ResNet-50, WideResNet, and AlexNet architectures. The experimental results are presented in Figure 7. As indicated by the accuracy curves, X-SAM consistently outperforms SAM, GSAM, and other baseline methods across all tested architectures and datasets, demonstrating superior generalization capabilities. Notably, we observed an intriguing phenomenon during our experiments: although X-SAM achieves the highest test accuracy, its corresponding test loss is not always the lowest; in some instances, it is marginally higher than that of standard SAM or other variants.

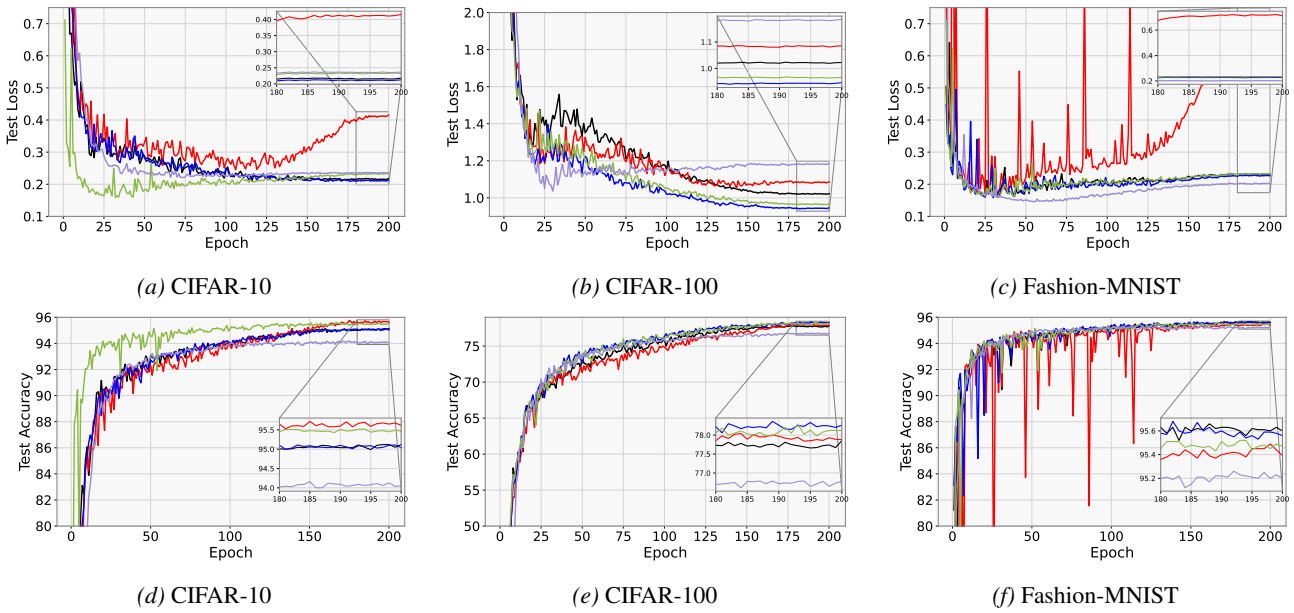
This phenomenon of "high accuracy accompanied by relatively higher loss" suggests that X-SAM does not merely pursue the minimization of cross-entropy loss (i.e., reducing the probability confidence error of predictions). Instead, it prioritizes the optimization of decision boundary robustness. A lower test loss often implies that the model is extremely "confident" (potentially over-confident) in its predictions, which can be an indicator of overfitting. Conversely, by explicitly penalizing high-curvature regions, X-SAM guides the optimization process into flatter minima. In such regions, the model exhibits more cautious predictions on misclassified samples, thereby avoiding extreme probability distributions on outliers. Consequently, the slightly elevated test loss actually reflects X-SAM's enhanced generalization ability and noise robustness, achieved while maintaining correct classification.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

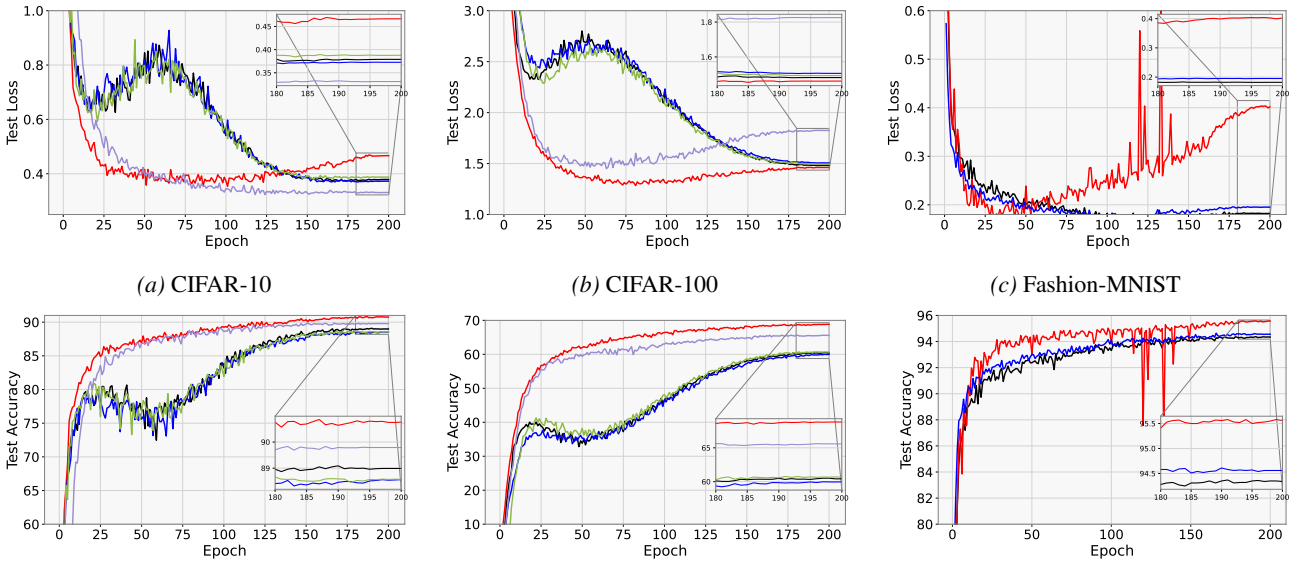
Figure 5. Learning curves for training ResNet-50 on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 5a, 5b and 5c: Test loss vs. Epochs; Figs 5d, 5e and 5f: Test accuracy vs. Epochs.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 6. Learning curves for training WideResNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 6a, 6b and 6c: Test loss vs. Epochs; Figs 6d, 6e and 6f: Test accuracy vs. Epochs.



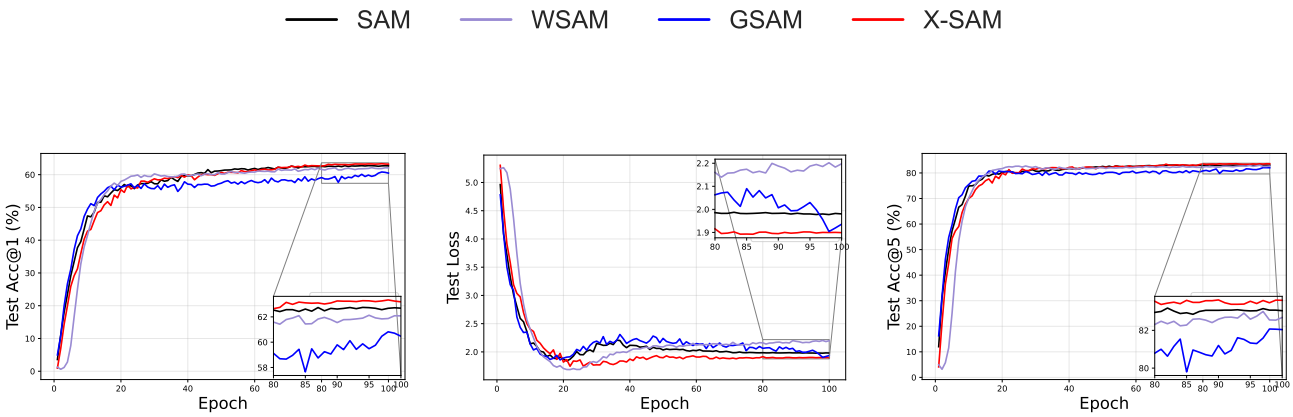
(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 7. Learning curves for training AlexNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 7a, 7b and 7c: Test loss vs. Epochs; Figs 7d, 7e and 7f: Test accuracy vs. Epochs.

C.2. Tiny ImageNet experiment

We evaluated the training dynamics of ResNet-34 on the Tiny ImageNet dataset, and the experimental results demonstrate that our proposed X-SAM exhibits exceptional generalization performance on this dataset. Within 100 training epochs, X-SAM achieved final test Top-1 and Top-5 accuracies of 63.34% and 83.61%, respectively, maintaining a significant and consistent lead over the standard SAM baseline (62.75% / 83.18%) as well as other variants such as GSAM and WSAM (see Fig. 8a and Fig. 8c). While all methods show rapid initial convergence followed by a steady ascent toward stabilization, X-SAM surpasses all competing schemes by a clear performance margin in the late training phase, as highlighted in the zoomed-in views of Fig. 8a. Simultaneously, the test loss curves in Fig. 8b indicate that X-SAM consistently maintains the lowest loss level throughout the entire training process, achieving deeper optimization while following a nearly identical convergence trajectory to the baseline. Overall, these results fully substantiate that X-SAM effectively boosts classification accuracy and generalization robustness on more challenging datasets like Tiny ImageNet.



(a) Top-1 Accuracy *(b)* Test Loss *(c)* Top-5 Accuracy

Figure 8. Top-1 Accuracy, Test Loss, and Top-5 Accuracy of ResNet-34 on the Tiny ImageNet dataset.

C.3. Robustness to Perturbation Radius

To investigate the effect of X-SAM on SAM's sensitivity to the perturbation radius ρ, we conducted experiments on the CIFAR-10 dataset using the ResNet-18 architecture. Specifically, we compared the test performance of SAM and X-SAM under different perturbation radii $\rho \in \{0.05, 0.1, 0.15, 0.2\}$, repeating each setting three times to report the averaged test accuracy. The figure 9 shows that as the perturbation radius increases, SAM's test accuracy initially decreases and then slightly recovers, whereas X-SAM maintains a generally stable performance. Even at $\rho = 0.1$, X-SAM preserves consistent generalization. These findings indicate that X-SAM alleviates SAM's sensitivity to the perturbation radius, enabling more stable training dynamics and consistent generalization across a wide range of ρ values.

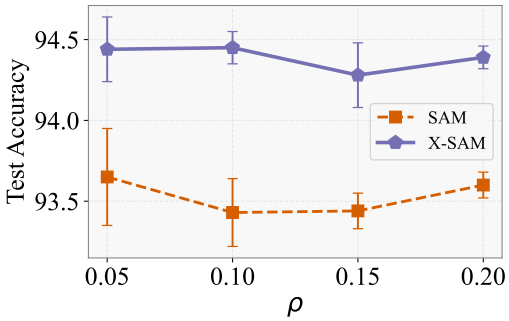


Figure 9. Comparison of test accuracy under different values of ρ on CIFAR-10 with the ResNet-18 model

C.4. Hyperparameter Sensitivity Investigation of α

We explore how the hyperparameter α influences the performance of X-SAM by running experiments on the CIFAR-100 dataset with the ResNet-18 architecture. Different values of α are evaluated, as illustrated in Figure 10. The results indicate that when $\alpha = 0.2$, the model achieves the highest test accuracy among all tested settings, showing a 1.07% improvement over standard SAM. This suggests that selecting an appropriate α can significantly enhance performance. Moreover, the test accuracy remains relatively stable when α varies within the range $[0.1, 0.4]$, indicating that α is a fairly robust hyperparameter.

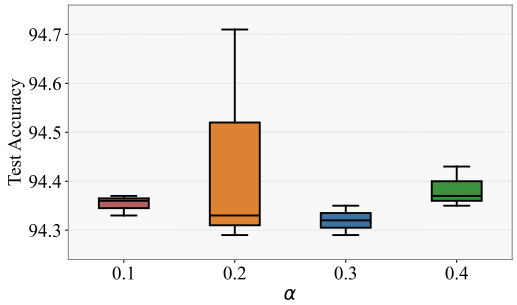


Figure 10. Test accuracy of ResNet-18 when training on CIFAR-10 under different values of α for X-SAM.

C.5. Hessian Spectrum Investigation

We analyze the impact of training with SAM and X-SAM on the Hessian spectrum of the model. The experiment is conducted on a ResNet-18 model trained on the CIFAR-100 dataset, where the top k eigenvalues of the Hessian matrix $\nabla^2 f(w)$ are computed at the second epoch and visualized. The results, as shown in Figure 11, show that compared with SAM, X-SAM consistently produces a smaller largest eigenvalue and a smaller trace, with a higher concentration of eigenvalues near zero. This indicates that X-SAM guides the optimization towards flatter regions of the loss surface, confirming the effectiveness of the proposed gradient correction mechanism and explaining why Eigen-SAM achieves better generalization performance than SAM.

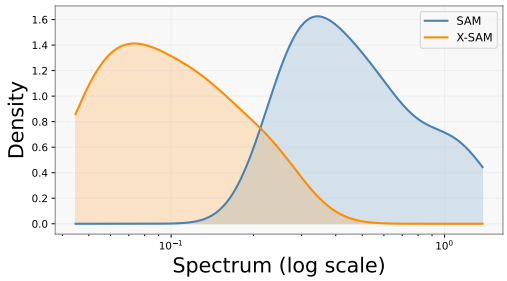


Figure 11. Hessian spectrum at the end of training ResNet-18 on CIFAR-100.

C.6. Hyperparameter Sensitivity Investigation of p

We explore how the hyperparameter of eigenvector update interval influences the performance of X-SAM by running experiments on the CIFAR-10 dataset with the ResNet-18 architecture. Different values of the update interval (denoted as freq) are evaluated, as illustrated in Figure 12. The results indicate that when freq = 3, X-SAM achieves the highest test accuracy of 94.36% among all tested settings, showing a 0.53% improvement over the worst-performing setting (freq=100, 93.83%). This suggests that selecting an appropriate eigenvector update interval can significantly enhance the performance of X-SAM. Moreover, the test accuracy of X-SAM remains relatively high and stable when the update interval is set to small values within [3,5] or the large value of 50, with all these settings outperforming the intermediate intervals of 10, 20 and 100, indicating that the eigenvector update interval is a fairly robust hyperparameter for X-SAM.

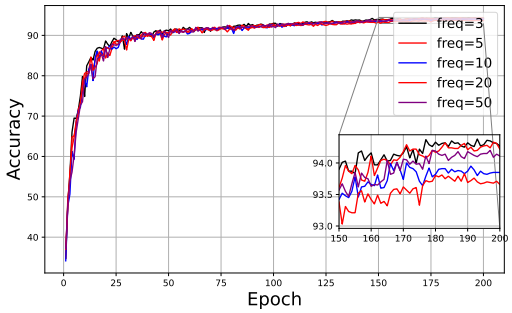


Figure 12. Test accuracy of ResNet-18 on CIFAR-10 with different eigenvector update intervals.

C.7. Hyperparameter Sensitivity Investigation of q

We explore how the hyperparameter of the number of power iterations influences the performance of X-SAM by running experiments on the CIFAR-10 dataset with the ResNet-18 architecture. Different values of power iteration steps are evaluated. The results, as illustrated in Fig. 13, show that X-SAM consistently outperforms the baseline SAM across all tested power iteration settings: X-SAM with 1, 3, and 5 power iteration steps achieves test accuracies of 94.39%, 94.43%, and 94.71% respectively, all of which are higher than the baseline SAM's accuracy of 93.76%. Specifically, when the number of power iterations is set to 5, X-SAM achieves the highest test accuracy among all tested settings. This suggests that increasing the number of power iterations can effectively enhance the performance of X-SAM, and selecting an appropriate number of power iterations is crucial for maximizing the performance gain of X-SAM.

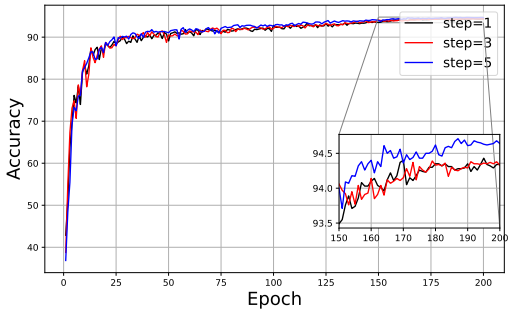


Figure 13. Test accuracy of ResNet-18 on CIFAR-10 with different numbers of power iterations.

C.8. ViT-Tiny model fine-tuning

We conduct a fine-tuning experiment on CIFAR-10 with the ViT-Tiny model under a fixed training budget of 4000 steps. As illustrated in Fig. 14, our proposed X-SAM achieves a test accuracy of 92.33%, which significantly outperforms the standard SAM baseline of 91.88%. This result verifies that X-SAM can effectively adapt to the complex non-convex attention loss landscape of vision transformers, demonstrating its strong cross-architecture generalization capability.

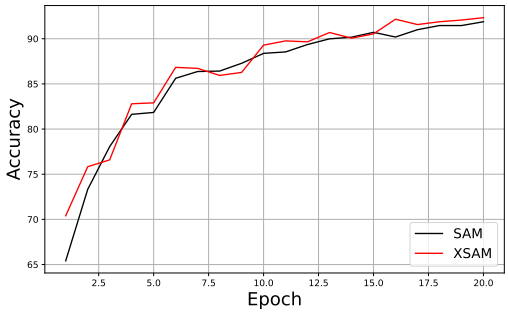


Figure 14. Test accuracy of the ViT-Tiny model fine-tuned on the CIFAR-10 dataset.

C.9. Test Accuracy under Equal Training Time

To evaluate the performance of various optimizers under an equivalent training time budget, we conducted comparative experiments on the CIFAR-10 dataset using the ResNet-50 architecture.

As illustrated in Fig. 15, test accuracy is visualized as a function of cumulative training time. The experimental results demonstrate that, given the same computational expenditure, both X-SAM and GSAM exhibit convergence efficiency significantly superior to that of the standard SAM baseline. Notably, while GSAM maintains a lead in the early training stages due to its rapid upward trend, X-SAM displays more sustained potential for performance improvement. As shown in the magnified inset, as training enters the mid-to-late stages (beyond approximately 6,000 seconds), the growth of GSAM gradually slows down and plateaus, whereas X-SAM maintains steady growth, eventually converging with or even slightly surpassing GSAM by the end of the training process. This phenomenon substantiates that while maintaining high convergence efficiency, X-SAM effectively prevents the model from reaching a premature performance bottleneck, achieving a superior trade-off between training efficiency and final accuracy.

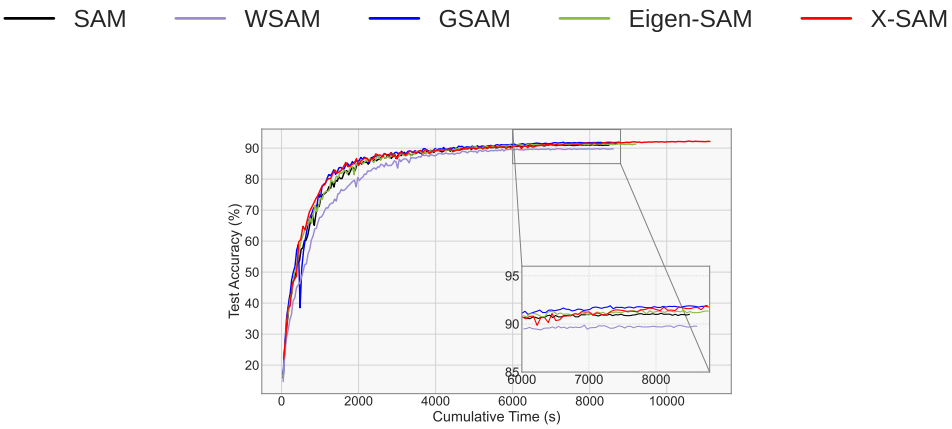
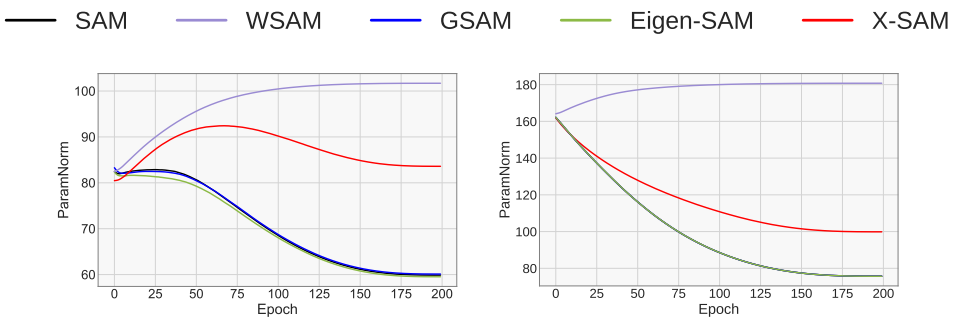


Figure 15. Test accuracy plotted against cumulative training time for ResNet-50 on the CIFAR-10 dataset.

C.10. Parameter Norm

We further analyze the parameter norm dynamics during training on the CIFAR-10 dataset, using both ResNet-18 and WideResNet-28 architectures. As shown in Fig. 16, our proposed X-SAM exhibits a nearly identical parameter norm evolution trend to the standard SAM baseline across both architectures. For ResNet-18, both methods follow a characteristic rise-then-decay pattern(see Fig. 16a), while for WideResNet-28, both show a consistent downward trend throughout training(see Fig. 16b). Notably, the parameter norm of X-SAM is consistently higher than that of SAM, with an approximate increase of 25% to 30% across the entire training process, which remains within a fully controllable range. This observation verifies that X-SAM inherits the stable optimization dynamics of SAM while maintaining a moderate, manageable increase in parameter norm, which contributes to its superior generalization performance without introducing excessive parameter

growth.



(a) ResNet-18 *(b)* WideResNet-28

Figure 16. Learning curves of parameter norms vs. epochs for training ResNet-18 and WideResNet-28 on CIFAR-10.

Table 2. Summarization of Test Accuracy (%) results. The best accuracy results are highlighted in bold.

Datasets	Methods	ResNet-18	ResNet-50	WideResNet	AlexNet
CIFAR-10	ASAM	94.23 ± 0.10	92.38	95.23	88.12
	X-SAM	$\mathbf{94.71 \pm 0.20}$	**93.83**	**95.75**	**90.79**
Fashion-MNIST	ASAM	94.91	94.29	**95.64**	94.24
	X-SAM	**95.58**	**95.09**	95.49	**95.58**

The symbol "–" indicates that the model failed to converge.

Table 3. Training Time per Epoch Across Different Models

Optimizer	ResNet-18	ResNet-34	ResNet-50	WRN-28-10	ViT-Base
SAM	12.33	22.25	42.90	79.49	312.24
X-SAM	15.15	27.35	55.85	95.47	380.06